Institute for Cognitive Systems
Technische Universität München
Prof. Gordon Cheng

Humanoid Sensors and Actuators
Project Definition

Summer Semester 2026

1 Project Description

You are free to propose your own final projects, but they should be related to the course and include the topics that we have worked with during the semester. In the next section, you will find some project directions that can serve you as guide for your proposals. For your final project, you can work in groups of **maximum 3 people** (2 people or individual projects are also allowed).

- **Thursday, 18.06.2026**: Project pitch presentation. Please submit to Moodle the following deliverables **(deadline is Thursday, 18.06.2026, 13:15 right before the lecture)**:

  1. Project description and task definition (1-2 pages): Explain your project in one paragraph and divide the tasks for it as detailed as you can. Assign the tasks to each member of your group. Bear in mind that part of the grading policies include the successful completion of your assigned tasks.
  2. Project pitch slides (max. 5 slides): These are the slides of your pitch presentation. Please be as concise as possible. **Your presentation should not take longer than 6 minutes (2 minutes per person).**
  3. List of materials: Submit a list of all the materials you will need for your project, quantities, prices, and the web address where to buy them. You can use the components we have worked with during the tutorial or order different/more advanced ones. The list of available vendors is:
     `https://eckstein-shop.de/`
     `https://www.conrad.de/`
     `https://www.mouser.de/`
     `https://de.rs-online.com/web/`
     `https://www.reichelt.de/`
     `https://www.berrybase.de/`
     `https://www.roboter-bausatz.de/`
     `https://www.extremtextil.de/`
     `https://www.fishing-adventure.com/`
     We have enough 3D printing material so you do not need to add this to your list. There are also different types of textile so please consult with us before adding it to your list. **We do not order from Amazon, eBay etc.**

- **Tuesday, 23.06.2026**: Feedback session with supervisors. After your project submission we will meet with each group to modify the projects and order the missing items.
- **Thursdays, 25.06.2026, 02.07.2026, 09.07.2026, 19.07.2026** Colloquium with supervisors to discuss project progress and preparation for the final presentation.
- **Thursday, 23.07.2026** Exam date. On this day you will present your project together with a demo and will submit the report and technical content related to it.

2 Project Directions

*Your project **should not** be a simple robotic project from the internet. You may base your ideas on them but if your proposal is too simple we will increase the workload to make it even among groups.*

As a starting point: Reflect on what you learned in the tutorials. Decide what was interesting for you; to what you would like to have a in-depth look.

Your project should contain sensors and actuators and ideally address humanoid robots.

- Building your own humanoid: Projects can include building parts of a humanoid or small size humanoid. Sensors and actuators need to be part of your project. Examples: Building smooth eyes for a humanoid, novel grip actuation (we have MR fluids in our lab that you can play with, see: here, and here)
- We are building soft actuators and soft sensors. Several types of materials can be used. You may use textile-coated TPU, silicon molding, etc. For actuation, we can offer you an air source, pneumatics valves, electronic elements, and DC motors. Please see the publications on
  - "High-Performance Perpendicularly-Enfolded-Textile Actuators for Soft Wearable Robots: Design and Realization"
  - "Soft pneumatic elbow exoskeleton reduces the muscle activity, metabolic cost and fatigue during holding and carrying of loads"
  - "Design of New Sensory Soft Hand: Combining Air-Pump Actuation with Superimposed Curvature and Pressure Sensors"

  For sensing projects, we can offer conductive materials such as conductive fabric, liquid metal, and iodide solutions. Please see the publications on

  1. "A Robust Data-Driven Soft Sensory Glove for Human Hand Motions Identification and Replication"
  2. "Multi-sensory fusion of wearable sensors for automatic grasping and releasing with soft-hand exoskeleton"

3 Project Grading Policy

Projects are graded according to the following criteria:

- **15% Presentation + Video (+ Demo)**
  - **Explain** your chosen engineering and scientific approaches.
  - **Elaborate** on what you designed, how you designed it, why your design was necessary, and what you conclude from your results or analysis.
  - Each group will have a time slot of maximum 8 min. Each student has to contribute for at least 2 minutes. For example if you are 3 students in a group, the group will have $\mathbf{3} \times \mathbf{2.5} \approx \mathbf{8}$ min in total to present the results, and if you are 2 students in a group, the group will have $\mathbf{2} \times \mathbf{2.5} + \mathbf{1} = \mathbf{6}$ min in total to present the results. The Q&A that follows will last at most 2 min.
- **30% Group Report**
  - Project reports should be written in a concise scientific style. That includes a comprehensive presentation style and structure. A confusing report of unconnected paragraphs, non-understandable descriptions, and lots of grammatical and spelling errors that distract the reader from the content you present, will have a negative impact in the grading of the report.
  - The report should present the **technical approach**:
    - Talk about the problems you encountered and the approaches proposed to solve them.
    - **Be specific**. Make the introduction concise: You do **not** need to explain details of data sheets or characteristics of components you used, unless you specifically exploit them in your approach.
  - Pictures can be included **provided that they bring some technical value to the report**. All pictures have to be referenced in the text and explained. A picture can support your explanations in the text but it can **not** replace them.
  - Depending on the number of pictures included, **each student** should **contribute 2 pages** (approx. 1000 words), but not more than 4 pages, to the report.
  - The contribution of each student has to be clearly highlighted and refer to the tasks (see the next pages for more details).
- **50% Code Technical Content & Project operation**
  - Each group member is assigned with a set of tasks to complete.
  - Please use a clear, well documented approach to integrate your tasks!
  - Coding tasks should be integrated into the project architecture. Failing to do so (i.e. providing a stand-alone piece of code, un-merged with the code of the whole project) will result in no points for the corresponding task unless the task is separate by nature.
  - Please note: Similar to the tutorials, you have to reference sources you used/modified and explain how they work. You must not solve your tasks by copy and pasting code from the internet. You need to implement/engineer your own solutions, especially if the implementation is the main contribution of your task. Plagiarism will lead to a zero point evaluation of the corresponding task. Missing contributions will also lead to a zero point evaluation.
- **5% Code Documentation**
  - Each code file should contain a header describing its main purpose and structure.

- Within each code file, each function should have a header defining its purpose, inputs and outputs.
- Within each function, the main processing steps should be documented.
- Do not forget to include a "`readme.md`" file describing the structure of the code project. Do not forget to include your project **dependencies** and how to install them.